WORK SHEET - 29 **SOT + TE + AUC+ DE**

Q.1 (A)

$$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \Rightarrow 2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \frac{1}{16}$$

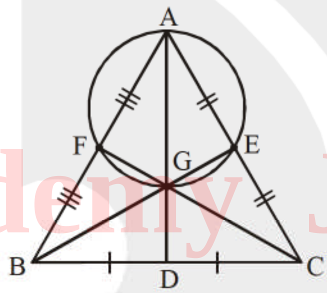
$$\Rightarrow \left(\cos\frac{A-B}{2} - \cos\frac{A+B}{2}\right)\sin\frac{C}{2} = \frac{1}{16} \qquad \left[\text{Taking into account } A - B = \frac{2\pi}{3}\right]$$

$$\Rightarrow \left(\frac{1}{4} - \sin\frac{C}{2}\right)^2 = 0 \qquad \left[\cos\left(\frac{A+B}{2}\right) = \sin\frac{C}{2}\right]$$

$$\Rightarrow \sin\frac{C}{2} = \frac{1}{4} \Rightarrow \cos C = 1 - 2\sin^2\frac{C}{2} = \frac{7}{8}$$ **Ans.**

Q.2 (C)

BG.BE = BF.BA



$$\Rightarrow \left(\frac{2}{3}BE\right)(BE) = \frac{c}{2}\times c$$

$$\Rightarrow \frac{2}{3}(BE)^2 = \frac{c^2}{2}$$

$$\Rightarrow \frac{2}{3}\left(\frac{2a^2 + 2c^2 - b^2}{4}\right) = \frac{1}{2}c^2$$

$$\left[\text{Length of median } BE = \frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2}\right]$$

$$\Rightarrow \frac{2a^2 + 2c^2 - b^2}{3} = c^2 \Rightarrow 2a^2 = b^2 + c^2$$ **Ans.**

Q.3 (B)

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\left.\begin{matrix} b_1 \\ b_2 \end{matrix}\right> b^2 - (2c\cos A)b + c^2 - a^2 = 0$$

$$\therefore\ b_1 + b_2 = 2c\cos A \quad \text{and} \quad b_1 b_2 = c^2 - a^2$$

$$\Rightarrow 3b_2 = 2c\cos A \quad \text{and} \quad 2b_2^2 = c^2 - a^2$$

$$2\left(\frac{2c\cos A}{3}\right)^2 = c^2 - a^2$$

$8c^2\cos^2 A = 9c^2 - 9c^2 \Rightarrow 8c^2(1-\sin^2 A) = 9c^2 - 9a^2$

$3a = c\sqrt{1+8\sin^2 A}$ **Ans.**

Q.4 (C)

$$\frac{2!}{10!}\,{}^{10}C_1 + \frac{2!}{10!}\,{}^{10}C_3 + \frac{1}{10!}\,{}^{10}C_5$$

$$= \frac{1}{10!}\left[\underbrace{{}^{10}C_1 + {}^{10}C_9} + \underbrace{{}^{10}C_3 + {}^{10}C_7} + \underbrace{{}^{10}C_5}\right] = \frac{1}{10!}2^{10-1} = \frac{2^9}{10!} = \frac{(2^3)^3}{10!} = \frac{(8)^3}{10!}$$

$a = 3,\ b = 5,\ c = 7$

$r = \frac{\Delta}{s}$, $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ **Ans.**

Q.5 (C)

$2\sin^3 x + 2\cos^3 x - 3\sin 2x + 2 = 0$

$\sin^3 x + \cos^3 x - 3\sin x\cos x + 1 = 0$

[Using : $A^3 + B^3 + C^3 - 3ABC = 0 \Rightarrow \frac{(A+B+C)}{2}\left((A-B)^2 + (B-C)^2 + (C-A)^2 = 0\right)$]

$$\left(\frac{\sin x + \cos x + 1}{2}\right)\left((\sin x - \cos x)^2 + (\sin x - 1)^2 + (\cos x - 1)^2\right) = 0$$

$\Rightarrow$ either $\sin x + \cos x + 1 = 0 \Rightarrow \sin x + \cos x = -1$

$\Rightarrow x = (2n+1)\pi$ or $x = (4k-1)\frac{\pi}{2}$

or $(\sin x - \cos x)^2 + (\sin x - 1)^2 + (\cos x - 1)^2 = 0$ has no solution.

So solution in $[0, 4\pi]$ are $\pi, 3\pi, \frac{3\pi}{2}, \frac{7\pi}{2}$; number of solutions is 4. **Ans.**

Q.6 (C)

Sol. $\cos y\, dx + (1 + 2e^{-x})\sin y\, dy = 0$

$$-\int \frac{\sin y}{\cos y}dy = \int \frac{dx}{1+2e^{-x}}$$

$\log(\cos y) = \log(e^x + 2) + C$

$$\frac{\cos y}{e^x + 2} = k$$

Put $x = 0$ and $y = \frac{\pi}{4} \Rightarrow k = \frac{1}{3\sqrt{2}} \Rightarrow e^x + 2 = 3\sqrt{2}\cos y$. **Ans.**

Q.7 (C)

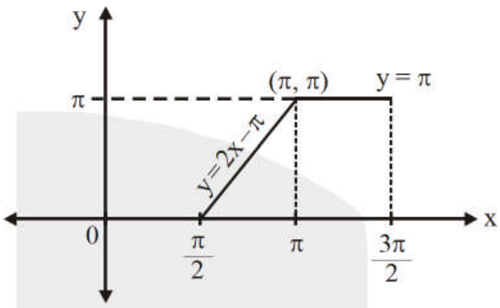
$$\frac{ye^x dx - e^x dy}{y^2} = y^2\,dy \Rightarrow d\left(\frac{e^x}{y}\right) = y^2\,dy \Rightarrow \frac{e^x}{y} = \frac{y^3}{3} + c$$

Put $y = 1$, $x = \ln\frac{1}{3}$, $c = 0$

Put $y = 3$, $x = \ln 27$ **Ans.**

Q.8 (B)

$f(x) = \tan^{-1}(\tan x) + \cos^{-1}(\cos x)$ for $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$



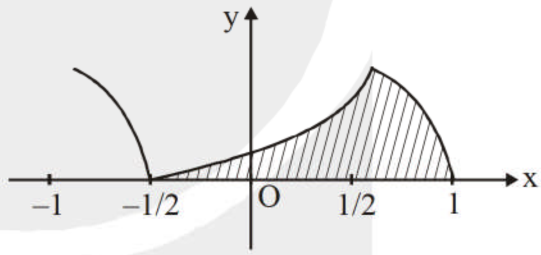
$$f(x) = \begin{cases} -(\pi - x) + x\ ; & \frac{\pi}{2} < x < \pi \\ (x - \pi) + (2\pi - x)\ ; & \pi \le x < \frac{3\pi}{2} \end{cases}$$

$$f(x) = \begin{cases} 2x - \pi;\ \frac{\pi}{2} < x < \pi \\ \pi\ ;\quad \pi \le x < \frac{3\pi}{2} \end{cases}$$

So required area $= \left(\frac{1}{2} \times \frac{\pi}{2}\right) \times \pi + \frac{\pi}{2} \times \pi = \frac{3\pi^2}{4}$. **Ans.**

Q.9 (A)

$$f(x) = \begin{cases} 3\cos^{-1}x, & x \in \left[\frac{1}{2}, 1\right] \\ 2\pi - 3\cos^{-1}x, & x \in \left[\frac{-1}{2}, \frac{1}{2}\right) \\ 3\cos^{-1}x - 2\pi, & x \in \left[-1, \frac{-1}{2}\right) \end{cases}$$



Area $= \int_{-1/2}^{1/2} \left(2\pi - 3\cos^{-1}x\right)dx + \int_{1/2}^{1} 3\cos^{-1}x\,dx = \frac{3\sqrt{3}}{2}$ sq. unit **Ans.**

$\left[\textbf{Note : } \int \cos^{-1}x\,dx = x\cos^{-1}x - \sqrt{1 - x^2}\right]$

Q.10 (A)

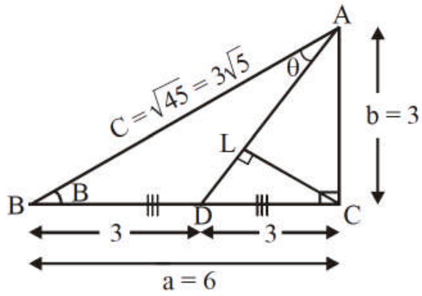
It solution lie on line $y = 3x$

$2\sin 3x + 8\sin^3 x = a$ should have real solutions

$\Rightarrow 2(3\sin x - 4\sin^3 x) + 8\sin^3 x = a \Rightarrow \sin x = \frac{a}{6}$

For no solution $\frac{a}{6} > 1$ or $\frac{a}{6} < -1$. **Ans.**

Q.11 (B)
Q.12 (C)
Q.13 (D)



$\cos(A-B)=\frac{4}{5}$, $\quad \sin(A-B)=\frac{3}{5}$, $\quad \tan(A-B)=\frac{3}{4}$

$$\frac{2\tan\left(\frac{A-B}{2}\right)}{1-\tan^2\frac{A-B}{2}}=\frac{3}{4} \Rightarrow 3\tan^2\frac{A-B}{2}+8\tan\left(\frac{A-B}{2}\right)-3=0$$

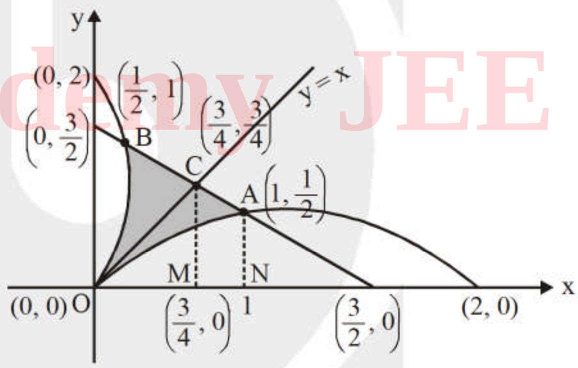
$$\Rightarrow \boxed{\tan\frac{A-B}{2}=\frac{1}{3}} \qquad \text{(As, } a>b) \qquad \text{[Using Napier's Analogy]}$$

$$\frac{a-b}{a+b}\cot\frac{C}{2}=\frac{1}{3} \Rightarrow \frac{6-3}{6+3}\cot\frac{C}{2}=\frac{1}{3} \Rightarrow \cot\frac{C}{2}=1 \Rightarrow C=90^\circ$$

$\therefore$ ΔABC is a right angled triangle. Length of median AD $=\sqrt{3^2+3^2}=3\sqrt{2}$

$$\frac{\sin\theta}{BD}=\frac{\sin B}{AD} \Rightarrow \sin\theta=\frac{3\sin B}{3\sqrt{2}}=\frac{3}{\sqrt{2}\times\sqrt{45}}=\frac{1}{\sqrt{10}}\text{. } \textbf{Ans.}$$

Q.14 [5]



$$f(x)=\frac{2x-x^2}{2}$$

Since $2x+2y=3$;

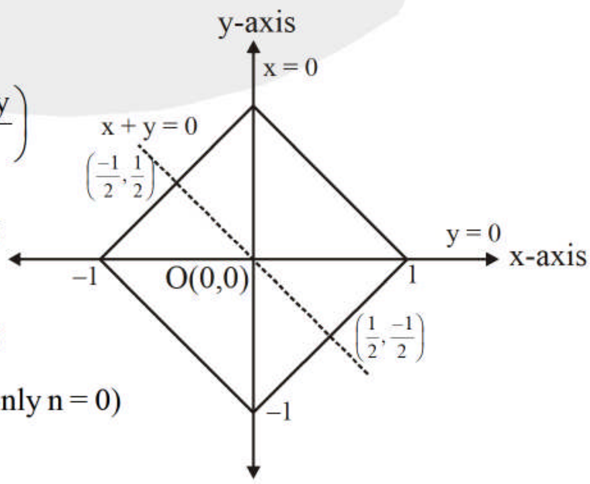
Passes through $A\left(1,\frac{1}{2}\right)$ and $B\left(\frac{1}{2},1\right)$.

So bounded area A = area OAB
= 2 [area OCM + area CMNA – area ONA]

$$=2\left[\frac{1}{2}\times\frac{3}{4}\times\frac{3}{4}+\frac{1}{2}\left(\frac{3}{4}+\frac{1}{2}\right)\times\frac{1}{4}-\frac{1}{2}\int_0^1(2x-x^2)\,dx\right]$$

$$=\frac{5}{24} \quad \Rightarrow 24A=5. \textbf{ Ans.}$$

Q.15 [0006]



$\sin x+\sin y=\sin(x+y)$

$$2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)=2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x+y}{2}\right)$$

$\Rightarrow \cos\frac{x+y}{2}=\cos\frac{x-y}{2}$ or $\frac{x+y}{2}=n\pi$

$\Rightarrow \frac{x+y}{2}=2m\pi\pm\left(\frac{x-y}{2}\right)$ | $x+y=2n\pi$

$\Rightarrow$ positive sign | $x+y=0$ (only $n=0$)

$y=2m\pi \Rightarrow y=0$ only

negative sign

$x = 2m\pi \Rightarrow x = 0$ only

$$(x, y) \in \left\{\left(\frac{1}{2}, \frac{-1}{2}\right), \left(\frac{-1}{2}, \frac{1}{2}\right), (1, 0), (-1, 0), (0, 1), (0, -1)\right\}.$$

Q.16 [0015]

$\tan 4x \cos 7x = \sin 7|x|$

$x \geq 0 \quad \sin 4x \cos 7x = \sin 7x \cos 4x$

$\sin(3x) = 0 \quad 3x = n\pi$

$x = \frac{n\pi}{3} \qquad n = 0, 1, 2, 3 \Rightarrow x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$ **(4 solutions)**

$x < 0 \quad \tan 4x \cos 7x = -\sin 7x \Rightarrow \sin 11x = 0 \Rightarrow x = \frac{n\pi}{11}$

$n = -1, -2, -3, \ldots$ upto (-11)

$\therefore x = \frac{-\pi}{11}, \frac{-2\pi}{11}, \frac{-3\pi}{11}, \ldots, \frac{-10\pi}{11}, -\pi$ **(11 solutions)**

Total 15 solutions Ans.

Note : We shall consider two cases :

Case-I : When $x \geq 0$

Case-II : When $x < 0$.

Q.17 [0003]

$y = x^2 \Rightarrow y = \sec^{-1}[-\sin^2 x]$

For $\sec^{-1} x$, $|x| \geq 1$

So, $\left|[-\sin^2 x]\right| \geq 1$

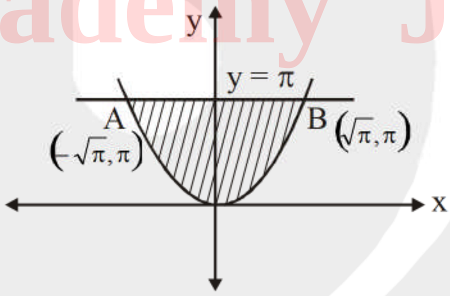
But $\sin^2 x \in [0, 1]$

$[-\sin^2 x] = 0, -1$

So, $y = \sec^{-1}[-\sin^2 x]$ is defined when

$= \sec^{-1}(-1) \qquad \sin^2 x \neq 0$

$= \pi \qquad x \neq n\pi$



$$\text{Area} = \int_{-\sqrt{\pi}}^{\sqrt{\pi}} (\pi - x^2)\,dx = \frac{4\pi\sqrt{\pi}}{3} = \frac{4(\pi)^{\frac{3}{2}}}{3}.$$ **Ans.**

Q.18 [0004]

$$\frac{dy}{dx} = y + \int_0^2 y\,dx \Rightarrow \frac{dy}{dx} = y + Q \Rightarrow \frac{dy}{y+Q} = dx \qquad \ln(y+Q) = x + P$$

Given $y(0) = 1 \qquad P = \ln(1 + Q)$

$$\ln\left(\frac{y+Q}{1+Q}\right) = x$$

$y + Q = (1 + Q)e^x$

$$Q = \int_0^2 \left((1+Q)e^x - Q\right)dx \Rightarrow Q = (1 + Q)(e^2 - 1) - 2Q$$

$$\Rightarrow 3Q = e^2 - 1 + Q(e^2 - 1) \Rightarrow Q = \frac{e^2-1}{4-e^2}$$

$$\therefore \quad y = \left(1 + \frac{e^2-1}{4-e^2}\right)e^x - \left(\frac{e^2-1}{4-e^2}\right) = \frac{3e^x - e^2 + 1}{4-e^2}$$

$$y(2) = \frac{2e^2+1}{4-e^2} \Rightarrow [|y(2)|] = 4. \textbf{ Ans.}$$

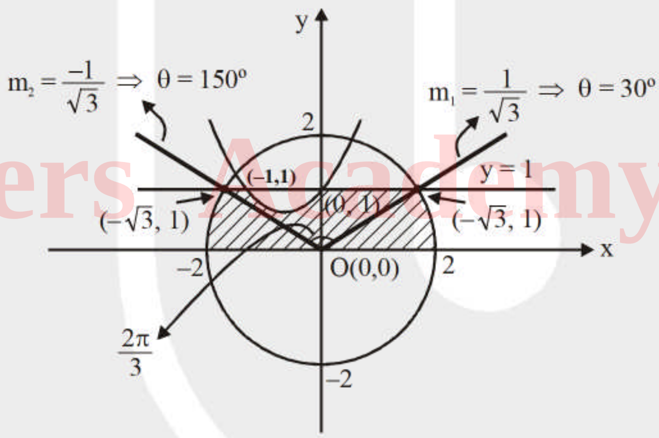
Q.19 [0006]

Since $\left[\sin^2\frac{x}{4} + \cos\frac{x}{4}\right]$ lies between 1 and 2 for $x \in [-2, 2]$

As, $\quad -2 \le x \le 2 \Rightarrow \frac{-1}{2} \le \frac{x}{4} \le \frac{1}{2} \subset \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$\therefore \quad \cos^2\frac{x}{4} \le \cos\frac{x}{4} \Rightarrow \sin^2\frac{x}{4} + \cos^2\frac{x}{4} \le \sin^2\frac{x}{4} + \cos\frac{x}{4} \Rightarrow \sin^2\frac{x}{4} + \cos\frac{x}{4} \ge 1$

thus the problem need as to be read as the area bounded by $x^2 + y^2 = 4$, $y = x^2 + x + 1$ and the line $y = 1$



Area of sector $= \frac{1}{2}r^2\theta = \frac{1}{2}(2)^2 \times \frac{2\pi}{3}$

Now, $\quad$ area $= \frac{1}{2}\pi \times 2^2 - \left(\frac{1}{2}\times 4 \times \frac{2\pi}{3} - \frac{1}{2}2^2 \times \sin\frac{2\pi}{3}\right) - \int_{-1}^{0}(1-(x^2+x+1))dx$

$$= 2\pi - \left(\frac{4\pi}{3} - 2\times\frac{\sqrt{3}}{2}\right) - \int_{-1}^{0}(1-(x^2+x+1))dx$$

$$= \sqrt{3} + \frac{2\pi}{3} - \frac{1}{6} \Rightarrow k = 6. \textbf{ Ans.}$$